

Oontecho

A rhythmic trance-gate multi-effect: a tempo-synced step gate that also moves the pan and a resonant filter, followed by a stereo delay and a reverb wash.



Overview

Oontecho is in the spirit of the classic rhythmic trance-gate multi-effects of the 2000s. At its heart is a step sequencer of up to 32 steps that chops the volume in time with the project. Each step has a level, drawn as a bar, and an accent. The same pattern can also throw the sound from side to side and play the cutoff of a resonant filter, so one pattern can gate, pan and sweep at once.

The signal runs GATE, then FILTER, then DELAY, then SPACE, then the master MIX and the output stage. The delay is a tempo-synced stereo echo with separate left and right times and a ping-pong mode; SPACE is a smooth eight-line reverb wash. Both add their sound on top of the gated signal, so a chopped pad can trail echoes and a tail into the gaps.

The gate follows the project. While REAPER plays it locks to the timeline, so it is in time from the very first beat; while the transport is stopped it keeps running freely at the project tempo, so you can audition sounds with it.

Reach for it for

- Classic trance-gated pads, strings and vocals
- Sidechain-style pumping without a sidechain
- Sequenced resonant filter lines from a sustained chord
- Stereo chops that bounce from side to side
- Dub-style gated throws into a dark ping-pong echo
- Slow, gliding swells into a long reverb wash

Controls

The header

PRESETS, ◀ ▶ and the display	PRESETS (or a click on the display) opens the list of factory presets; ◀ and ▶ step through them, and the display shows the one in use. A new instance opens on Init.
Tempo display	The project tempo in BPM. The light and the word SYNC show while REAPER plays and the gate is locked to the timeline; FREE shows while it is stopped and the gate runs on at the project tempo.
DRIFT	0 to 100% (25% by default). See Drift below.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window. A new instance opens in Large.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

The step grid

The pattern itself, across the top of the panel. Each column is one step; the bar is its level and the cell below is its accent. The beats are shaded in alternate tones, and the step that is playing is outlined in cyan.

STEPS 8 / 12 / 16 / 24 / 32	The length of the pattern (16 by default).
DIV	The length of each step: 1/4, 1/8, 1/8 T, 1/16, 1/16 T or 1/32 (1/16 by default). 16 steps of 1/16 is one bar.

< / >	Rotate the pattern one step left or right, within the current length. Use them to move the groove against the beat.
RND	Makes a random pattern on all 32 steps: every fourth step is full, the others are a mix of rests and half, three-quarter and full hits, with a few accents.
FILL	Sets all 32 steps to full and clears the accents: a blank canvas.
Level bars	Click and drag across the bars to draw the levels, 0 to 100%; a quick sweep paints a smooth line across the steps it passes. Right-drag erases (sets steps to 0). Hover over a step to see its level. Changes are heard at once, even on the step that is playing.
Accent row	Click a cell to switch that step's accent on or off; drag along the row to paint the same state across several steps. Accented steps are drawn in amber.

GATE

The PATTERN MODULATION knobs: what the pattern does to the sound.

DEPTH	How far the pattern moves the volume, 0 to 100% (100% by default). At 100% a silent step is silent; at 0% the volume is left alone and the pattern only drives PAN and FILTER.
GLIDE	How smoothly one step moves to the next, 0 to 100% (15% by default). At 0% the edges are fast but click-free, for a hard chop; higher settings soften them, up to a slow swell over half a step at 100% - the classic pumping feel.
ACCENT	How much the accented steps stand out, shown as 0 to +6 dB (+3 dB at the default 50%). Accented steps are louder by that much and also open the filter by up to 1.5 octaves.
SWING	Delays every second step, 0 to 50% (0% by default). At 50% the off-step lands half a step late.

PAN	-100 to +100% (0% by default). The pattern moves the sound across the stereo field: odd steps to one side, even steps to the other, each as far as its level. With + the first step goes left; - flips the sides.
FILTER	-5 to +5 octaves (0 by default). The pattern plays the cutoff of the FILTER section: a full step moves it this far above (+) or below (-) CUTOFF, a silent step leaves it at CUTOFF. Only heard when a filter type is chosen.

FILTER

A smooth, resonant state-variable filter after the gate.

OFF / LP / BP / HP	The filter type: off, low-pass, band-pass or high-pass (OFF by default). Changing type crossfades, so it never clicks. The CUTOFF cap turns yellow for HP.
CUTOFF	The filter's base frequency, 20 Hz - 20 kHz (2 kHz by default). The pattern, the accents, the LFO and DRIFT all move it from here.
RESO	Resonance, 0 to 100% (20% by default). Higher settings give a sharper, more vocal peak; the level is trimmed a little as it rises.
LFO	The depth of a built-in sine LFO on the cutoff, -4 to +4 octaves (0 by default). Negative values start the sweep the other way.
RATE	The LFO's cycle, from 4 bars down to 1/32, including dotted (D) and triplet (T) values (1 bar by default). It locks to the timeline while REAPER plays.

DELAY

A tempo-synced stereo echo, after the filter. The display shows the time and the left and right multipliers.

PING	Ping-pong: the echoes start on the left and bounce from side to side. Off by default.
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TIME	The echo time as a note value, from 4 bars down to 1/32, including dotted (D) and triplet (T) values (1/8 D by default). Echoes are limited to 4 seconds.
L / R	Multiply the time on each side: x0.5, x0.75, x1, x1.5 or x2 (x1 by default). The knob shows the resulting time in ms. Different values on each side give a wide, rhythmic spread.
FDBK	Feedback, 0 to 95% (35% by default): how many repeats you hear. The loop is soft-limited, so it stays musical when pushed.
DAMP	A high-cut in the feedback loop, 1 - 20 kHz (6 kHz by default): lower settings make each repeat darker than the last. A fixed low cut in the loop keeps the repeats from building up mud.
MIX	The level of the echoes on top of the gated sound, 0 to 100% (0% by default, so the delay is silent until you raise it).

SPACE

A smooth reverb wash after the delay, so the echoes are reverberated too. The display shows the decay time in seconds.

SIZE	The size of the space, 0 to 100% (50% by default).
DECAY	How long the tail lasts, 0.4 to 10 seconds (0 to 100%; 50%, about 2.8 seconds, by default). Longer settings are also a little darker.
MIX	The level of the reverb on top, 0 to 100% (0% by default).

MASTER

TRIM	Input level before anything else, -24 to +12 dB.
GUARD	Click the TRIM knob (without turning it) to switch GUARD on or off; the small light at the knob's upper right shows it is on and glows brighter while it works. Off by default.

	Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS. It never boosts quiet sources.
SAT	Band saturation on the way out, 0 to 100% (0% by default): only the lows and highs are saturated, the mids stay clean.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper at -0.3 dBFS (no latency), or a lookahead limiter (3 ms latency, compensated by REAPER). CLIP is on by default and flashes red while it clips.
MIX	Dry / wet balance of the whole effect, 0 to 100% (100% by default). Lower it to keep some of the ungated sound underneath.
OUTPUT	Output level, -24 to +12 dB.
OUT meter	Left and right output levels.

Drift

DRIFT is never rhythmic: the timing always stays locked to the grid. Instead the filter cutoff wanders slowly (up to about a sixth of an octave either way at the default 25%, just over half an octave at 100%), and the delay's colour breathes with it - its damping and a little of its feedback. At 0% everything stays exactly where you set it.

Knobs: drag up or down (Shift for fine moves); double-click to reset a knob to its default. The mouse wheel works too, and the value shows while you hover.

Presets

Presets set the pattern, STEPS, DIV and every GATE, FILTER, DELAY and SPACE control, and the master MIX. OUTPUT, DRIFT and the TRIM / GUARD, SAT and output stage are left as they are.

Init	16 full steps of 1/16 with no accents, filter, delay or space: a blank canvas to draw on.
Classic 16ths	The 3-3-2 trance chop in 1/16ths with accents, dotted-eighth echoes and a little space.

Euphoric Gate	Accented 1/16ths through a resonant low-pass at 900 Hz that the pattern opens, with echoes and a wider space.
Sidechain Pump	Ducked on every beat and swelling back up, with plenty of glide: sidechain pumping without a sidechain.
Half-Time Chop	Long 1/8 hits across two bars, a gentle low-pass and a quarter-note echo behind them.
Filter Sequence	The volume stays put; the pattern plays the cutoff of a very resonant low-pass from 160 Hz, up to 4.5 octaves.
Dub Delay Gate	Sparse 1/8 hits through a high-pass, feeding a dark dotted-quarter ping-pong echo with long feedback.
Ambient Wash	A slow, gliding 8-step quarter-note gate with a two-bar LFO on a low-pass, a long echo and a large, dark space.
Triplet Trance	One bar of 1/16 triplets (24 steps), rolling across the stereo field, with triplet echoes.
Ping-Pong Pan	1/16ths that fly hard left and right, with ping-pong echoes bouncing behind them.
32nd Stutter	A bar of 1/32 chops that roll into the next downbeat, with a short 1/16 echo.
LFO Sweep Gate	A swung 1/16 gate through a resonant band-pass swept three octaves by the LFO over two bars.

Tips

- Put it on a sustained source - a pad, a held chord or a long vocal note. The gate needs something to chop.
- The FILTER knob in GATE, ACCENT's filter lift and the LFO do nothing while the filter type is OFF: choose LP, BP or HP first.
- For a pure filter sequence, turn DEPTH to 0 and FILTER up: the volume stays even while the pattern plays the cutoff.

- Raise GLIDE for a pumping, sidechain-style feel; keep it low for a hard, choppy gate.
- Delay and space MIX are added on top, so the gaps in the pattern fill with echoes and tail. Raise them slowly - a little goes a long way.
- Use the < and > keys to try the same pattern starting on a different step; it can change the groove completely.

File: gfx-oontecho.jsfx